

AS BUSINESS - DAY 8

Capacity and quality

Utilisation, inventory and quality control

YOUR AIM TODAY

Calculate capacity use and analyse operational trade-offs.

YOUR RULE

Read the notes twice. Use exact case facts. Every explanation must contain a because link and a therefore business impact. Keep page 4 closed.

Student: Rohan

Start time: _____ Finish time: _____

Date: _____

Knowledge and method

Capacity is the maximum output possible in a period.

- Capacity utilisation=(actual output/maximum output)x100.
- High utilisation spreads fixed cost but leaves little room for surges or maintenance.
- Inventory protects against shortages but ties up cash and may become obsolete.
- Quality control inspects output; quality assurance builds quality into the process.
- Poor quality creates waste, returns and reputational damage.

MINI CASE

BakeBox can make 10,000 snack boxes monthly and currently makes 9,200. Defect rate is 6%. A supermarket order needs another 1,200 boxes next month.

WORKED CHAIN

Capacity utilisation= $9200/10000 \times 100 = 92\%$. The extra order exceeds spare capacity of 800, so BakeBox needs overtime, outsourcing or extra capacity.

Knowledge, application and analysis

Use the mini case in Questions 3-6.

Question	Your answer
1. Define capacity utilisation.	
2. Calculate BakeBox's utilisation.	
3. Calculate spare capacity.	
4. Explain one risk of accepting the order immediately.	
5. Explain one benefit of quality assurance.	
6. Recommend one operational response.	

QUALITY CHECK

Circle every case fact. Underline every because link. Box every business impact.

Models and correction log

Wording may differ, but the knowledge and chains must be accurate.

Q	Answer / model points
1	Actual output as a percentage of maximum possible output.
2	92%.
3	$10,000 - 9,200 = 800$ boxes.
4	Required extra output is 1,200, above spare capacity; rushed overtime may worsen the 6% defect rate and harm the supermarket relationship.
5	Preventing errors during production can reduce waste and returns, lowering cost and protecting reputation.
6	Use limited vetted outsourcing or temporary capacity only if quality standards can be monitored; otherwise negotiate a phased delivery.

Q	Missing layer	My improved sentence

SEND FOR MARKING

Send pages 3 and 4. Ask for a level, missing chains, and a rewritten model for the weakest response.